



Discussion 9

Filesystems, Reliability

04/12/24

Staff

Announcements

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		RPC Lab Deadline			Project 3 Design Doc Due	
				HW5 Checkpoint Deadline		
		HW5 Due		Midterm 3		

Reliability

Availability

Availability is the probability that the system can accept and process requests.

- Measured in “nines” of probability (e.g. 99.9% = “3-nines of availability”)
- Maximize independence of failures

Durability

Durability is the ability of a system to recover data despite faults (i.e. fault tolerance)

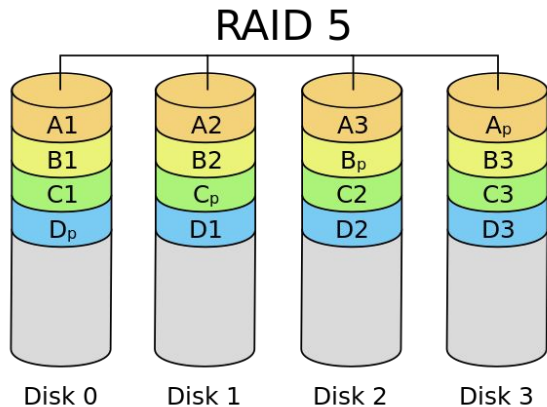
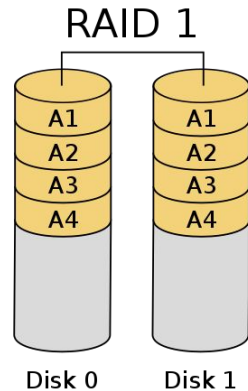
- *Durability does not necessarily imply availability*

Small defects (e.g. single disk failure) → use Reed-Solomon error correcting codes.

Short term data preservation (e.g. buffer cache, delayed writes) → use special, battery-backed RAM (NVRAM) for buffer cache.

Long term data preservation → use Redundant Array of Inexpensive disks (RAID)

- Spread data redundantly across multiple disks to tolerate individual disk failures.
- **RAID 1** will **mirror** a disk onto another “shadow” disk.
- **RAID 5** stripes data across n multiple disks to allow for *single* disk failure.
 - Each **stripe unit** consists of $n-1$ blocks and one **parity block**, which is created by XOR-ing the $n-1$ blocks.
 - If a disk fails, simply XOR the remaining disks.



Reliability

Reliability the ability of a system or component to perform its required functions under stated conditions for a specified period of time.

- Includes availability, durability, security.

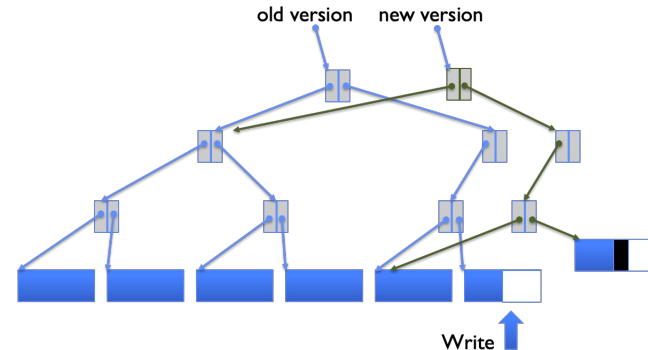
Careful ordering and recovery

- Break down an operation into smaller chunks.
- **Not a foolproof method but works most of the times.**

Copy on write (COW)

- Write new versions of data into a new place on disk instead of updating in place.
- Update mappings and references to data last.
- **Easy recovery if system crashes in the middle of updating data since old data is fully intact.**
- **Data is only being added, not modified → batch updates, parallel writes**

Creating a File	
Careful Ordering (FFS)	Recovery (fsck)
1. Allocate data block.	1. Scan inode table.
2. Write data block.	2. If any unlinked files (not in any directory), delete or put in lost and found directory.
3. Allocate inode.	3. Compare free block bitmap against inode trees.
4. Write inode block.	4. Scan directories for missing update/access times.
5. Update free map.	
6. Update directory entry.	
7. Update modify time for directory entry.	



Concept Check

1. What benefit with regards to read bandwidth might you see from using RAID 1?
2. What is the minimum number of disks needed to use RAID 5?
3. RAID 4 had a dedicated disk with all the parity blocks. On the other hand, RAID 5 distributes the parity blocks across all disks in a round robin fashion. Why is this approach beneficial in terms of write bandwidth?
4. How can COW help with write speeds?

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Two copies of same data → doubled read bandwidth

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All parity blocks on one disk → this disk becomes the bottleneck for all writes.

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Can transform random I/O into sequential I/O.

